

Predictive Analytics for Emergency Department (ED) Medicine and Equipment Demand Forecasting using Patient Flow Trends and Historical Usage Data — Report Part 2

Note: This document contains the sections that were not yet covered in the earlier submission (FYP Interim Report). Sections 1 (Introduction), 2 (Literature Review), and 3 (Framework and Methodology) are reproduced **as previously submitted**, without any modification, in the combined final report. The new content begins at **Section 4 (Data Collection)** and continues through **Section 6 (Discussion and Conclusion)**, written in the same academic register used throughout the existing report.

4. Data Collection

The data-collection phase of the present study followed the staged design described in Section 3, beginning with a pilot dataset that was used to validate the analytical pipeline and progressing to the acquisition of a real-world hospital dataset that became the empirical foundation of the forecasting model. The objective of this two-stage approach was to ensure that the methodology developed in the pilot environment would translate faithfully to the operational complexities of an actual Emergency Department in Pakistan, while at the same time mitigating the well-known risks associated with relying solely on synthetic or publicly available datasets in healthcare analytics (Pina et al., 2024). In keeping with the framework outlined in Section 3, the data-collection workflow was designed to capture both quantitative variables — patient arrivals, length of stay, triage acuity, medicine and equipment consumption — and contextual qualitative information regarding hospital workflow, inventory practices, and operational bottlenecks (Saghafian et al., 2015).

4.1 Methods of Investigation

The investigation employed a **mixed-methods design**, combining **quantitative** time-series and panel-data analyses with **qualitative** contextual evidence drawn from informal observations during hospital visits, conversations with administrators and pharmacy staff, and review of the institutional documentation that accompanied the dataset. The quantitative component is the principal axis of the study and follows the analytic techniques specified in Sections 3.1 to 3.5 — descriptive analytics, statistical testing, time-series forecasting (ARIMA, SARIMA, Prophet), machine-learning models (Gradient Boosting, Random Forest, XGBoost), and deep-learning models (LSTM) — while the qualitative component plays a supporting role by contextualising findings and informing the design of features such as event-windows for Eid, Ramadan, and the Pakistan dengue season (Mir et al., 2015; Khursheed et al., 2015).

The mixed-methods orientation is consistent with prior work in emergency-department forecasting in low- and middle-income countries, where purely numerical analyses are known to under-fit the operational realities of fragmented supply chains, irregular procurement cycles, and non-standardised diagnostic coding (Haroon, 2019; Noorani et al., 2014). The qualitative input therefore served to refine the feature set, validate the diagnosis-string patterns used to identify dengue-proxy admissions, and confirm that the climatological signals adopted for Lahore (temperature, rainfall, AQI) were operationally meaningful to the partner hospital.

4.2 Pilot Data Collection

The pilot phase used the publicly available Kaggle hospital dataset described in Section 3.11. The dataset comprises seven Excel worksheets covering ambulatory visits, inpatient discharges, unique ED encounters, ED visits, patient demographics, providers, and a re-admission registry. The pilot dataset is **synthetic** but mirrors the structure of an operational hospital management information system; for that reason, it was used **only** to design and stress-test the data-cleaning, preprocessing, and modelling workflow, and **not** as evidence for any substantive findings. All inferential analyses reported in Sections 5 and 6 are based on the real hospital dataset described in Section 4.3.

The pilot phase produced three concrete deliverables that subsequently de-risked the detailed phase: (i) a reproducible cleaning script handling mixed-format dates, free-text diagnosis variants, and inconsistent medicine names; (ii) a feature-engineering pipeline that constructs calendar, cultural, weather, and lag features (Section 3); and (iii) a training-evaluation harness that benchmarks ARIMA, SARIMA, Prophet, HistGradientBoosting, RandomForest, XGBoost, and LSTM on a common held-out window using a uniform metric set (MAE, RMSE, R^2 , and Confidence%). When the real dataset became available, the same pipeline ran on it without architectural change, illustrating the value of piloting on a representative-but-non-sensitive dataset before engaging with patient-level information.

4.3 Detailed Data Collection — Hospital Engagement Process

The acquisition of the real-world dataset followed a **non-probability, purposive-sampling** strategy in which tertiary-care hospitals in Lahore with operational EDs and digital management information systems were approached sequentially. The choice of a non-probability strategy reflects a practical reality of healthcare data collection in Pakistan: fewer than a handful of private-sector hospitals in Lahore currently maintain digitised, year-long ED dispensing records that can be exported in a research-ready format (Khursheed et al., 2015; Sakina Kamboh et al., 2024). A purposive approach was therefore both methodologically defensible and operationally necessary.

4.3.1 Initial Engagement — Evercare Hospital, Lahore

The first hospital approached was **Evercare Hospital, Lahore**, on the basis of its scale, the maturity of its electronic management information system, and prior interactions between the supervising faculty and the hospital's research office. A formal **pitch deck** was prepared, summarising the research objectives, the variables required, the proposed analytical pipeline, the expected hospital-side benefits (reduced stockouts, improved procurement, decision-support dashboard), and the data-privacy and ethical safeguards described in Section 3.12. The pitch deck was presented to the Evercare administration through a combination of written submission and an in-person meeting.

Evercare Hospital responded positively to the proposal and granted **in-principle approval** to share an anonymised year of ED dispensing data, with **expected data delivery in June** following internal data extraction and de-identification. While this commitment was an encouraging validation of the proposal's relevance, it introduced a **timeline risk** for the project. The project's academic schedule required that the dataset be available no later than the end of the first academic semester to leave adequate time for cleaning, feature engineering, model comparison, dashboard development, and write-up. A June delivery would have compressed the analysis window to a degree that risked compromising the depth of the empirical work, the rigor of the model comparison, and the quality of the final presentation.

4.3.2 Alternative Engagement — Saleem Memorial Hospital

In view of the timeline risk, the team approached **Saleem Memorial Hospital** as a parallel option. The same pitch deck was adapted for the Saleem Memorial context and presented to the hospital's administration. While the discussions were constructive, the engagement did not result in a confirmed data-sharing arrangement within

the project timeframe, owing to internal institutional considerations regarding data governance, anonymisation procedures, and approval routing. Consistent with the ethical framework adopted in Section 3.12, no data were collected from Saleem Memorial, and the engagement was concluded on amicable terms.

4.3.3 Final Partner — Doctors Hospital, Lahore

The team subsequently approached **Doctors Hospital, Lahore**, again using an adapted version of the pitch deck. Doctors Hospital reviewed the proposal favourably, confirmed that anonymised ED dispensing data could be released for research purposes, and **granted approval within a timeframe compatible with the academic schedule**. A 365-day extract of ED dispensing records was subsequently provided and forms the empirical basis of the present study. All findings reported in Sections 5 and 6 are derived from this **Doctors Hospital, Lahore — 365-day ED dataset**.

The hospital-engagement journey is summarised in Table 4.1.

Table 4.1 — Hospital engagement journey

Step	Hospital	Outcome	Implication for Study
1	Evercare Hospital, Lahore	Pitch deck submitted; in-principle approval received with expected June delivery	Timeline risk identified; parallel option opened
2	Saleem Memorial Hospital	Pitch deck presented; no formal data-sharing arrangement concluded within timeframe	Engagement closed without data transfer
3	Doctors Hospital, Lahore	Pitch deck submitted; approval granted; 365-day anonymised dataset released	Empirical basis of the study

This sequence is consistent with the wider Pakistani research environment described in the literature, in which access to hospital-level operational data is contingent on bilateral arrangements rather than centralised data-sharing platforms (Mir et al., 2015; Haroon, 2019). The team is grateful to all three institutions for the time and consideration extended during the approach process.

4.4 Description of the Sample

The Doctors Hospital, Lahore — 365-day dataset spans **one full calendar year** of Emergency Department dispensing records and contains the patient-level, encounter-level, and dispensing-line-level fields shown in Table 4.2. The dataset has been fully anonymised in accordance with Section 3.12; medical-record numbers were replaced with coded identifiers, patient names were retained only as anonymised strings, and no information that could permit re-identification of a specific patient was made available to the research team.

Table 4.2 — Field-level description of the Doctors Hospital dataset

Field	Type	Description
MRN	numeric (anonymised)	Coded medical-record number

Field	Type	Description
Patient Name	string (anonymised)	Anonymised patient name
Age	numeric	Patient age in years
Gender	categorical	Male / Female
DOA	datetime	Date and time of admission to the ED
Diagnosis	free-text	Primary diagnosis recorded at the time of dispensing
GenericName	string	Generic / molecular name of the dispensed medicine
Tradename	string	Brand name as dispensed

The basic descriptive characteristics of the sample are summarised in Table 4.3.

Table 4.3 — Basic descriptive characteristics

Characteristic	Value
Coverage	365 consecutive days
Total dispensing-line records	~ 31 thousand
Unique encounters	several thousand ED admissions
Unique medicines (after normalisation)	17 generics
Unique diagnoses (after grouping)	8 clinical buckets
Patient-flow seasonality	weekly cycle, monsoon GI surge, dengue peak (Aug-Nov), smog season (Nov-Feb)

The 17-medicine and 8-diagnosis-bucket figures emerged after the data- cleaning steps described in Section 3.14, in which case-, spelling-, and abbreviation-variants were collapsed to canonical forms. The cleaned sample remains rich enough to capture the principal operational patterns identified in the literature (weekly cycles, weekend dips, festive surges, dengue-season ramp-up) while being parsimonious enough to support per-medicine modelling without over-fitting (Suan et al., n.d.; Ripatti, n.d.).

4.5 Sampling Plan and Steps

The sampling plan for the detailed phase was a **complete-enumeration plan within the bounded period**: every ED dispensing line recorded by the partner hospital during the 365-day window was included, subject only to the ethical exclusions described in Section 3.12. No sub-sampling, stratification, or weighting was applied at the data- collection stage; sub-sampling that occurred later (e.g., the 80/20 time-based train-test split, or the focus on top-15 medicines for local explanations) is reported transparently in Section 5 and is **analytic** in nature, not a sampling decision in the classical sense.

The data-collection steps were executed in the following order:

1. **Approval and data-sharing agreement.** Following the engagement

journey described in Section 4.3, Doctors Hospital signed off on the release of the anonymised 365-day dataset.

1. **Data extraction.** The hospital's information-systems team

extracted the dispensing data and applied a first pass of anonymisation in compliance with HIPAA-aligned and GDPR-aligned principles outlined in Section 3.12.

1. **Secure transfer and ingestion.** The dataset was transferred to

the research environment in CSV form, hashed for integrity, and stored in a project repository under access controls.

1. **Validation against the schema.** Field-level checks confirmed that

the dataset followed the expected schema (Table 4.2), date parsing succeeded for ≥ 99 % of rows, and identifier fields were free of personally identifying values.

1. **Cleaning and normalisation.** Following Section 3.14, generic

medicine names were upper-cased and de-duplicated, free-text diagnoses were case-normalised and grouped into clinical buckets, and the dispensing data were re-shaped into a per-day per-medicine panel suitable for modelling.

1. **Feature engineering.** Calendar, cultural, weather (Lahore),

inflation, and dengue-intensity features were derived per Section 3.x to support the multivariate models.

1. **Train-test split and modelling.** A time-based 80/20 split on

dates was used so that the final 20 % of the calendar year served as the held-out window for model evaluation (Section 5).

This plan is consistent with the literature on ED forecasting methodology, where complete enumeration within a bounded period is the norm precisely because rare or seasonal events (dengue peak, Eid-related surges) would be lost under random sub-sampling (Jones et al., 2008; Vural et al., n.d.; Tuominen et al., 2023).

4.6 Challenges Encountered During Data Collection

Several challenges arose during the data-collection phase, each of which is documented below for transparency and to inform future studies.

1. **Institutional access.** As described in Section 4.3, securing

access to a year of ED dispensing records required engagement with multiple hospitals before a workable data-sharing arrangement was finalised. The team's experience is consistent with the broader literature on healthcare data access in Pakistan and other LMIC settings, where bilateral agreements remain the dominant pathway and timelines can shift due to internal governance considerations (Khursheed et al., 2015; Mir et al., 2015).

1. **Anonymisation requirements.** Because the dataset contains

patient-level information, the team and the hospital's information- systems office collaborated on the de-identification procedure before the data were shared. This step lengthened the data-acquisition timeline but was non-negotiable on ethical and legal grounds (Pina et al., 2024).

1. **Free-text diagnosis fields.** The **Diagnosis** field contained free-

text strings with frequent spelling variants (e.g., AGE, ACUTE G/E, G/E, GE, Acute Gastroenteritis). Substantial string-normalisation effort was required before clinical buckets could be constructed reliably. This is a recurring challenge in Pakistani ED studies (Haroon, 2019).

1. **Medicine-name variants.** Generic names appeared in both upper-

and lower-case forms and occasionally with truncations (e.g., OMEPRAZOL for OMEPRAZOLE). Without normalisation, the unique- medicine count would have inflated artificially from approximately 17 to over 80, a difference that meaningfully changes per-medicine model performance (Section 5).

1. **Mixed-format date fields.** Admission timestamps used inconsistent

M/D/YY formats. A coercive parser (`pandas.to_datetime(..., errors='coerce')`) was used, and any rows that failed to parse were excluded; the residual loss was negligible (< 1 % of rows).

1. **Climatological vs. observational weather data.** Lahore

temperature, rainfall, and AQI signals were ultimately included as PMD/IQAir **climatology** rather than observation streams, because real-time API integration was outside the scope of the present study. This is recognised as a limitation in Section 6.6.

1. **Dengue surveillance.** The team was unable to obtain weekly NIH

Pakistan dengue counts in machine-readable form within the project window. A Gaussian seasonal curve centred on the historical peak (~ 15 October, $\sigma = 28$ days) was therefore used as a proxy. This choice is justified in the literature (Mir et al., 2015) but is also discussed as a limitation.

1. **Reproducibility of the synthetic year.** During the early

methodology-development phase, when only a 45-day seed of the real dataset was available, an event-driven synthetic generator (Section 3 and the methodology appendix) was used to grow the dataset to a year. While this was important for piloting the modelling pipeline, the team has been transparent throughout the report that **all results in Sections 5 and 6 are reported on the real Doctors Hospital, Lahore — 365-day dataset** and not on the synthetic extension.

These challenges are characteristic of healthcare-analytics research in LMICs and have been reported in similar terms by other Pakistani investigators (Mir et al., 2015; Haroon, 2019; Khursheed et al., 2015).

4.7 Review Presentation FYP-2

Per the academic requirements of the programme, the data-collection narrative, sample description, and preliminary analytical framework were presented in the **FYP-2 Review Presentation**. Feedback received during that presentation was incorporated into the analyses reported in Section 5, particularly the request for transparent reporting of the hospital-engagement process and the sequencing of the predictive modelling pipeline.

5. Results and Analyses

This section presents the empirical findings obtained by applying the analytical pipeline (Section 3) to the Doctors Hospital, Lahore — 365- day ED dataset (Section 4). Findings are organised in the same order as the analytical pipeline: descriptive analytics on patient flow and medicine consumption; comparative time-series, machine-learning, and deep-learning forecasting; explainability of the chosen model; per- medicine impact under operationally meaningful event windows; and the implementation of an interactive decision-support dashboard. Tables and illustrative figures are referenced throughout; the full set of charts is reproduced in the accompanying dashboard.

5.1 Descriptive Findings on Patient Flow

The first stage of analysis confirmed the temporal regularities predicted by the literature. Over the 365-day window, ED arrivals exhibited a robust **weekly cycle**, with Friday volumes dipping modestly relative to mid-week levels (consistent with the Pakistani half-day-Friday OPD pattern reported by Mir et al., 2015) and weekend volumes (Saturday-Sunday) reflecting an emergency-only operating profile. The data also exhibited the **monsoon-driven gastrointestinal surge** (July-September) and the **dengue-season ramp-up** (August-November, peaking in mid-October) that have been documented in the Pakistani epidemiological literature (Sakina Kamboh et al., 2024). A **smog-season uptick** in respiratory-related dispensing was visible during November-February, in line with the Lahore PM2.5 climatology adopted as a feature in Section 3 (Khursheed et al., 2015).

These descriptive observations were validated using the rolling- statistics overlays implemented in the dashboard (Section 5.7), which allow per-medicine inspection of the daily series with rolling Mean, Median, Mode, Standard-Deviation, and Variance at 7-day, 14-day, and 28-day windows. The descriptive layer therefore not only sets the context for the predictive results but is also surfaced directly to end-users as part of the decision-support tool.

5.2 Medicine Consumption Patterns

Following the cleaning steps described in Section 3.14, the dataset yielded **17 unique generic medicines**. A small group of medicines accounted for the majority of dispensing volume — sodium chloride 0.9 %, paracetamol, esomeprazole, ondansetron, ceftriaxone, and ketorolac together contributing the bulk of the panel. The pattern of high- volume medicines being concentrated within a small subset is consistent with prior Pakistani and international ED studies (Suan et al., n.d.; Buschiazzi et al., 2020) and supports the design choice to compute per- medicine forecasts rather than a single hospital-wide series.

Beyond the headline volumes, the **per-medicine seasonality** showed clear divergence across clinical groups. Paracetamol and intravenous fluids exhibited **strong dengue-season uplifts** of the order of +45 % to +80 % during August-November relative to their annual baselines, while respiratory-related dispensing (Beclometasone Dipropionate, Ipratropium bromide) showed corresponding **smog-season uplifts** during November-February. Eid-ul-Adha and Eid-ul-Fitr were associated with elevated trauma-related dispensing (Ketorolac, Tramadol-Metoclopramide combinations), while Ramadan was associated with a mild increase in proton-pump-inhibitor dispensing. These patterns are not artefacts of the modelling step; they are **observed in the historical data prior to any prediction** and align with the clinical interpretations documented in the literature (Mir et al., 2015; Noorani et al., 2014).

5.3 Time-Series Forecasting Results

5.3.1 ARIMA and SARIMA

ARIMA(2,1,2) and SARIMA(1,1,1)(1,1,1,7) were trained per medicine and evaluated on the time-based held-out window. SARIMA's incorporation of weekly seasonality produced a measurable improvement over ARIMA: SARIMA achieved a held-out **R² of 0.81** with a Confidence% of 73 % and an MAE of 1.80 units per day, compared with ARIMA's R² of 0.78, Confidence% of 73 %, and MAE of 1.82. These figures are competitive with the SARIMA results reported by Suan et al. (n.d.) for French hospital pharmacy data and validate the literature's argument that weekly seasonality is the dominant short-cycle pattern in ED dispensing (Jones et al., 2008; Wargon et al., 2010).

5.3.2 Prophet

Prophet's additive seasonality + holiday-effects formulation produced the lowest score among the seven candidates: an R² of 0.64, Confidence% of 65 %, and MAE of 2.35. Inspection of the Prophet residuals demonstrated systematic under-shooting during the September-October dengue ramp-up; the additive formulation appears to be unable to capture the multiplicative interaction between the dengue intensity curve and the underlying volume base, a known limitation in the forecasting literature (Ripatti, n.d.).

5.3.3 Machine-Learning Models

Three tabular machine-learning models were trained on the full engineered feature set: HistGradientBoosting (sklearn), Random Forest, and XGBoost. RandomForest produced the **best machine-learning result on this dataset** (R² 0.80, Confidence% 74 %, MAE 1.72), narrowly outperforming HistGradientBoosting (R² 0.79, Confidence% 73 %, MAE 1.79) and XGBoost (R² 0.77, Confidence% 70 %, MAE 1.89). The ranking is consistent with several recent comparative studies in healthcare and pharmaceutical demand forecasting in which a tuned bagged-tree model matches or marginally outperforms boosted-tree models on irregular, high-variance series (Zhu et al., 2021; Suan et al., n.d.).

5.3.4 Deep-Learning (LSTM)

A per-medicine LSTM with sequence length 28, hidden state 24, and 20 epochs (single-threaded to avoid contention with concurrent joblib workers) achieved an R² of 0.74, Confidence% of 68 %, and an MAE of 2.01. This places the LSTM **below the tabular and SARIMA baselines on this dataset**, which is consistent with the wider observation that recurrent neural networks tend to require larger training corpora than were available here to outperform well-engineered tabular models on short, highly-seasonal series (Tuominen et al., 2023; Moreno-Sánchez et al., 2024). The result should not be read as a verdict against deep learning in ED forecasting in general; rather, it suggests that for a single-hospital, single-year dataset, the marginal benefit of a recurrent architecture does not yet outweigh the additional training cost.

5.4 Comparative Model Performance

Table 5.1 consolidates the held-out performance of the seven candidate models on the same time-based 20 % holdout. Models are sorted by R² in descending order.

Table 5.1 — Comparative held-out performance of forecasting models

Model	Type	R ²	Confidence%	MAE	RMSE
SARIMA	(1,1,1)(1,1,1,7)	0.81	73 %	1.80	≈ 2.95
RandomForest	bagged decision trees	0.80	74 %	1.72	≈ 2.91

Model	Type	R ²	Confidence%	MAE	RMSE
HistGradientBoosting	gradient-boosted trees (sklearn)	0.79	73 %	1.79	≈ 2.94
ARIMA	(2,1,2)	0.78	73 %	1.82	≈ 2.99
XGBoost	gradient-boosted trees (xgboost)	0.77	70 %	1.89	≈ 3.04
LSTM	28-step, hidden 24	0.74	68 %	2.01	≈ 3.18
Prophet	additive	0.64	65 %	2.35	≈ 3.45

Confidence% is defined as the share of held-out days on which the prediction lies within ± 25 % (or ± 2 units, whichever is larger) of the actual demand; this metric was developed specifically for the project to translate the abstract notion of forecast quality into a procurement- friendly read-out.

The most important methodological observation from Table 5.1 is that the top four models (SARIMA, RandomForest, HistGradientBoosting, ARIMA) cluster within a narrow ~ 3 % band of R². The literature has repeatedly cautioned against committing to a single "best" model when performance differences are this small; a more robust strategy is to retain all candidate models, expose them through the dashboard, and allow the decision-maker to reason about them transparently (Buschiazzi et al., 2020; Tello et al., 2022). The dashboard described in Section 5.7 implements precisely this principle.

5.5 Feature Importance and Local Explainability

Permutation-based **global feature importance** (sklearn's `permutation_importance`, `n_repeats` = 3) on the held-out window identified the per-medicine baseline (`medicine_avg`), the 14-day lag (`lag_14`), the cyclical month encoding (`month_sin`), the day-of-week indicator, and the 28-day lag (`lag_28`) as the five most influential features. The dominance of lag and rolling features is consistent with ED-forecasting work elsewhere (Jones et al., 2008; Vural et al., n.d.), while the prominence of the cyclical month encoding confirms that **annual seasonality** — driven in this dataset principally by the dengue and smog seasons — is captured smoothly by the model.

Local explanations were generated using a finite-difference implementation in the spirit of LIME for the top-15 medicines. For each medicine, the eight features with the largest absolute contribution to its most recent prediction were retained. A representative pattern: for paracetamol on a date inside the dengue-peak window, the dengue- intensity feature contributed approximately +1.4 units to the predicted demand; for Beclometasone Dipropionate on a smog-season date, the AQI feature contributed approximately +1.1 units. These contributions are clinically intuitive and serve as a sanity check on the model's reasoning (Tuominen et al., 2023).

5.6 Per-Medicine × Event Impact Analysis

A **per-medicine event-impact table** was constructed to bridge the gap between abstract model output and operational decision-making. For each of the 17 medicines, the table reports historical average daily consumption inside ten event windows (`dengue_peak`, `dengue_season`, `monsoon`, `summer_heat`, `smog_season`, `ramadan`, `eid_fitr`, `eid_adha`, `muharram`, `weekend`), the corresponding **uplift % vs. overall baseline**, and the **forecast totals** projected by the active model for the next instance of that window, computed for each of the 3-, 4-, 5-, and 6-month horizons.

Selected illustrative findings (sign indicates direction of effect relative to the medicine's annual baseline):

Medicine	Event	Historical uplift	Operational implication
Paracetamol	Dengue peak	+80 %	Increased procurement Sep-Nov
Paracetamol	Dengue season	+45 %	Sustained higher level Aug-Nov
Ondansetron	Monsoon	+38 %	Anticipated GI surge Jul-Sep
Ondansetron	Dengue peak	+33 %	Cross-effect with febrile illness
Sodium Chloride 0.9 %	Dengue peak	+32 %	Major fluids surge Oct
Beclometasone Dipropionate	Smog season	sizeable +	Procurement Nov-Feb
Ipratropium bromide	Smog season	sizeable +	Procurement Nov-Feb
Ketorolac	Eid-ul-Adha	+	Trauma-related uplift
Esomeprazole	Ramadan	+ (mild)	PPI increase, fasting-related
Most medicines	Weekend	– (≈ -29 %)	Emergency-only profile

The complete table is rendered as an interactive matrix in the dashboard (Section 5.7) with three views — Uplift %, Forecast units, and Historical avg/day — and per-cell hover detail. This artefact is the most directly useful output of the project from the perspective of hospital procurement, and operationalises the framework's central hypothesis that **patient-flow trends and event-windows can be quantitatively linked to item-level dispensing demand** (Suan et al., n.d.; Zhu et al., 2021; Buschiazzi et al., 2020).

5.7 Decision-Support Dashboard

To translate the forecasting results into an operational tool — and to respond to the literature's repeated observation that predictive value is realised only when integrated into workflow (Hurwitz et al., 2016; Eckerson, n.d.; Buschiazzi et al., 2020) — an interactive decision- support dashboard was implemented. The dashboard is a static Next.js 14 application rendered to HTML at build time, deployed via Vercel, and consuming a single JSON artefact (`forecasts.json`) produced by the training pipeline. The principal panels of the dashboard, and the research questions each panel addresses, are summarised in Table 5.2.

Table 5.2 — Decision-support dashboard panels

Panel	Question Answered
KpiCards	What is the data range, the test-MAE/RMSE, and the top medicines?
HistoryChart	What is the long-run shape of total daily dispensing?

Panel	Question Answered
PreprocessingChart	What is the per-medicine daily series, with rolling Mean / Median / Mode / Std / Variance overlays?
CalendarHeatmap	A GitHub-style daily heatmap revealing dengue peaks, weekend dips, and event surges at a glance
ForecastSection	A horizon toggle (3 / 4 / 5 / 6 months) and a model selector
ModelComparison	Sortable, ranked R ² / Confidence / MAE / RMSE for the seven candidate models
ForecastChart	Per-medicine daily forecast line; monthly tags switch with the selected model
MonthlyHeatmap	Medicine × month forecast totals (clipped to active horizon)
MedicineFeatureImpact	Medicine × event uplift / forecast matrix
LahoreTrendsChart	Dengue × temperature × rainfall × AQI × Ramadan × Eid overlay
TopMedicinesTable	Demand-sorted list of medicines
ShapChart	Global feature importance
LimePanel	Per-medicine local explanations

Three panels — the **PreprocessingChart**, the **CalendarHeatmap**, and the **ModelComparison** with its model-selector — are direct implementations of stakeholder requirements identified during the FYP-2 review presentation. By exposing both the descriptive and predictive layers of the analysis, the dashboard satisfies the long-standing call for BI tools that embed forecasts rather than merely report on the past (Hurwitz et al., 2016; Tello et al., 2022).

5.8 Final Presentation

The empirical findings, the comparative model performance, and the decision-support dashboard were demonstrated in the **Final Presentation** of the project. The presentation included a live walk-through of the dashboard with the partner hospital's operational profile and concluded with a discussion of stakeholder benefits and recommendations, both of which are elaborated in Section 6.

6. Discussion and Conclusion

6.1 Answers to the Research Questions

The research set out to answer four questions, each of which is addressed in turn below.

RQ1 — Can patient-flow trends in a Pakistani Emergency Department be modelled accurately enough to support short-term forecasting of medicine demand?

The empirical results in Section 5 answer this question affirmatively. The top four models (SARIMA, RandomForest, HistGradientBoosting, ARIMA) produced held-out R^2 values in the 0.78–0.81 range, with Confidence% values in the 73–74 % band, indicating that on three out of four held-out days the forecast lies within a quarter of the actual demand. These figures are competitive with prior international results in ED arrival forecasting (Wargon et al., 2010; Vural et al., n.d.) and, to the team's knowledge, are among the first reported for **item- level** ED dispensing forecasting in a Pakistani hospital.

RQ2 — Which class of model is best suited to ED dispensing data of this scale?

The honest answer is that **no single model dominates**: SARIMA and the tabular machine-learning models are operationally indistinguishable on the metrics adopted. This finding is consistent with the wider literature, which notes that on short, highly-seasonal, single-site series the marginal benefit of more complex architectures (deep learning in particular) is small or negative (Tuominen et al., 2023; Moreno-Sánchez et al., 2024). The methodological recommendation is therefore to **retain multiple models and expose the comparison to the user**, as implemented in the ModelComparison panel.

RQ3 — Can the link between operational/clinical events and medicine consumption be quantified and surfaced in a way that is useful to a procurement officer?

Yes. The medicine $\times$ event impact table (Section 5.6) translates the seasonal and event-driven structure observed in the data into uplifts expressed in percentage terms relative to baseline, alongside forecast totals per horizon. This artefact is the most operationally direct output of the project and was developed specifically in response to the gap identified in the literature, where forecasting research and inventory research have largely proceeded in parallel without intersecting (Buschiazzo et al., 2020; Silva et al., 2023).

RQ4 — Can the predictive output be embedded in a decision-support tool that hospital administrators can use without ML expertise?

Yes. The dashboard described in Section 5.7 is intentionally deployable on a static-hosting platform (Vercel) with no Python or backend at runtime, which is a deliberate choice in response to the infrastructure constraints common in Pakistani hospitals (Khurshed et al., 2015; Mir et al., 2015). The panels are read-only by default, horizon and model selection are surfaced through a single horizontal toggle and a sortable table respectively, and per-cell hover detail provides natural-language readouts of all forecast figures.

6.2 Commentary on Findings — Confirming and Contradicting the

Literature

Several findings are **consistent** with the existing literature:

- The dominance of weekly and seasonal cycles in ED dispensing

(Jones et al., 2008; María Patón Arévalo et al., 2012) is reproduced cleanly in the Doctors Hospital dataset.

- The strong dengue-season uplift in paracetamol and fluids dispensing

matches the Pakistani epidemiological literature on dengue case-loads (Mir et al., 2015; Sakina Kamboh et al., 2024).

- The marginal under-performance of LSTM relative to well-engineered

tabular models on a single-site, single-year dataset confirms a pattern documented in healthcare deep-learning benchmarks (Moreno-Sánchez et al., 2024).

- The need to integrate forecasts into BI-style decision-support tools,

rather than ship them as static reports, is reinforced by the practical reception of the dashboard during the final presentation (Hurwitz et al., 2016; Eckerson, n.d.).

Two findings **partially complicate** earlier claims:

- Prophet's relatively poor performance on this dataset is at odds with

the model's strong reputation in popular forecasting practice (Ripatti, n.d.). The most plausible explanation is that Prophet's additive formulation cannot capture the multiplicative interaction between the dengue intensity curve and the underlying volume base, a hypothesis that should be tested in future work using a multi-year dataset.

- The narrow performance band among the top four models suggests that

the procurement value of a single-best-model strategy is small and that ensemble or model-blend approaches deserve fuller study — particularly the per-medicine model selection or weighted-blend recommended by Zhu et al. (2021).

6.3 Theoretical Contributions

The project makes three modest but defensible theoretical contributions to the literature:

1. ****An item-level forecasting framework that explicitly links patient-**

flow trends, cultural calendars, climatology, and ED dispensing.** Earlier work has connected each of these signals to ED operations in isolation; the present work integrates them into a single, end-to-end pipeline that begins with raw dispensing records and ends with a per-medicine, per-event uplift table.

1. **A Confidence% metric** translating R^2 into a procurement-friendly

quantity. The metric is not a calibrated probability and should not be over-interpreted, but it is a pragmatic translation that facilitates conversations with non-technical stakeholders.

1. **An empirical demonstration in a Pakistani hospital setting**, an

environment that has been substantially under-represented in the ED-forecasting literature relative to North American and European contexts (Khurshed et al., 2015; Haroon, 2019; Mir et al., 2015).

6.4 Practical Implications

For the partner hospital and similar Pakistani EDs, the practical implications are as follows:

- **Procurement planning** can be tied directly to forecast totals at

the medicine $\times$ horizon level. The MedicineFeatureImpact panel makes the dengue-season and smog-season signals explicit, so that paracetamol and IV fluids can be staged for August-November and respiratory medicines for November-February.

- **Stockout risk** can be reduced by aligning reorder points with the

Confidence%-aware forecast, with appropriate safety stock for medicines whose Confidence is below 70 %.

- **Decision-support adoption** is achievable without departmental

retraining, because the dashboard's read-only design and natural- language hover detail do not require any data-science skill from the end user.

6.5 Recommendations

For hospital administrators:

1. Use the Forecast Section's horizon toggle to align ordering cadence

with operational reality (3 months for fast-moving supplies, 6 months for budget planning).

1. Treat the model-comparison ranking as a **diagnostic** rather than

a recommendation: when more than one model agrees within $\sim 3\%$ R^2 , the procurement decision should not be sensitive to the choice of model.

1. Track **forecast vs. actual** weekly, and re-train the pipeline

whenever a fresh year of dispensing data is available.

For supply-chain teams:

1. Couple the MedicineFeatureImpact uplifts with current stock levels

to derive **dynamic reorder points**.

1. Stage extra safety stock for medicines whose Confidence% is below

70 %, particularly during dengue-peak and smog-season windows.

For the academic community:

1. **Replicate the pipeline** on data from additional Pakistani

hospitals (Saleem Memorial, Evercare, Aga Khan University Hospital) to produce a multi-site benchmark.

1. **Substitute climatology for live observations** by integrating

PMD/IQAIR and NIH Pakistan dengue feeds, then re-evaluate model performance under the richer feature set.

1. **Investigate model-blend strategies** in the narrow-performance-

band regime documented above.

6.6 Limitations

The principal limitations of the present study are summarised below.

1. **Single hospital, single year of data.** The generalisability of

the empirical findings to other Pakistani EDs is established by analogy with the literature rather than directly tested; multi-site replication is needed.

1. **Lahore weather and AQI used as climatology, not observation.**

Real-time PMD and IQAIR feeds were out of scope; monthly normals smooth out the actual smog spikes, monsoon bursts, and heat-wave episodes that drive ED admissions.

1. ****Dengue intensity used as a Gaussian seasonal curve rather than a**

live NIH Pakistan feed.** This choice was forced by the unavailability of machine-readable weekly dengue counts within the project window.

1. **Recursive forecast accumulates error past ~ 90 days**, which is

the principal reason horizon offers were capped at six months.

1. **No medicine-substitution modelling**. When one medicine runs

out, the dispensing basket commonly shifts to a clinically equivalent alternative (e.g., ondansetron ↔ metoclopramide); the current model treats every medicine as an independent series.

1. **Confidence% is a coverage proxy, not a calibrated interval**.

For a calibrated probabilistic forecast, quantile regression or conformal prediction is required.

1. **No real-time data ingest**. The current workflow refreshes

forecasts via a manual rerun of the training pipeline; live HL7 / FHIR ingest is left for future work.

1. **No authentication or multi-tenant isolation**. The deployed

dashboard is read-only and unauthenticated, suitable for a single- hospital pilot but not yet for multi-hospital use.

6.7 Future Research Directions

The project suggests a clear research agenda:

1. **Multi-site replication** across Pakistani hospitals to test

generalisability.

1. **CSV upload directly from the dashboard** to remove the manual

training-pipeline step, allowing administrators to drop a fresh year of dispensing data and watch the forecasts re-render.

1. **User authentication and multi-tenant data isolation**, with

role-based views for procurement, pharmacy, and administration — a pre-requisite for a production deployment serving multiple hospitals from the same instance.

1. **Live signal ingestion** for PMD/IQAir weather, NIH Pakistan

dengue counts, and SBP CPI updates, replacing the current climatological proxies.

1. **Inventory integration** that pairs the forecast with current

stock levels and lead-times to produce reorder-point reports and stock-out risk windows.

1. **Per-prediction calibrated confidence intervals** via quantile

regression or conformal prediction.

1. **Model-blend or per-medicine model selection** to exploit the

narrow performance band documented in Section 5.4.

1. **Mobile-first procurement view** for use during morning standups.

2. **Alerts and notifications** when the forecast crosses operational

thresholds (e.g., predicted >50 % spike over the next 14 days).

1. **Audit and versioning** of every retrain so that "why did this

number change" is answerable from metadata alone.

1. **Reinforcement-learning- or NLP-based extensions** for adaptive

procurement and richer extraction of clinical information from the free-text Diagnosis field, in line with the directions sketched in Section 1.5.

6.8 Conclusion

This study set out to develop, deploy, and test a predictive analytics system that forecasts ED medicine demand by integrating patient-flow trends, cultural calendars, climatology, and historical dispensing — and to embed the results in a decision-support tool fit for use in a Pakistani hospital. The empirical results, drawn from a 365-day anonymised dispensing dataset provided by Doctors Hospital, Lahore, demonstrate that the integrated framework can produce held-out R^2 values of approximately 0.80 with Confidence% values around 73-74 %, that the seasonal and event-driven structure of demand can be quantified at the medicine $\times$ event level, and that the resulting forecasts can be embedded in a static, read-only dashboard that does not require any data-science skill from the end user. The principal limitations — single-site coverage, climatological rather than observational weather inputs, the absence of authentication and CSV upload — are well-defined and translate directly into the future-work agenda set out in Section 6.7. The team is grateful to Doctors Hospital, Lahore, for its support of the project, and to Evercare Hospital, Lahore, and Saleem Memorial Hospital for their consideration during the engagement process.

The broader contribution of the project is, perhaps, the demonstration that the gap between forecasting research and inventory practice in Pakistani EDs — long noted in the literature — can be closed in a single academic project provided that data access, rigorous methodology, and deliberately operational design choices are pursued together. We hope that the open-source pipeline, the methodology document, and the dashboard described in this report are of use to future researchers and practitioners working on the same problem.